Synthetic Dataset & Ground-Truth Validation

The Short Version

Every claim on this project's other pages — the 41%100% recall jump, the sigma threshold results, the indexing bugs — was only provable because the test data came with a known-correct answer attached. This page covers the dataset itself, and why that one property mattered more than the dataset's size.

What the Generator Produces

A version-controlled, seeded script produces a reproducible 7-day, 10,000-event dataset in one command. "Seeded" means the same seed always regenerates the exact same dataset — useful for re-running a test and trusting the data didn't quietly change underneath it.

Every event is labeled with one of three cohorts, decided by the generator itself at creation time, not inferred afterward:

- **normal** — realistic mixed traffic: Q&A, coding questions, summarization requests
- **drift** — sessions that start normal and pivot partway through into a prompt-injection or jailbreak attempt
- **noise** — random, non-semantic gibberish, included mainly as a control

Why the Label Matters More Than the Data

The label is assigned *before* the text is written — the generator picks "this row will be drift," then writes matching attack-style text, then stores the label. This makes `cohort` ground truth: a fact about the data that's true by construction, not a guess made by looking at it later.

That distinction is what turns "does this look about right" into an actual measurable claim. Without labels, 10,000 rows of search results can look plausible even when they're wrong — a wrong answer doesn't look like an error, it looks like *an* answer. With labels, you can ask a precise question: of the sessions truly labeled `drift`, how many did the detector actually catch? That question is what caught the IVFFlat empty-index bug, and it's what produced every recall and false-positive number in this project's other pages.

What This Enabled, Concretely

- **Catching the IVFFlat bug**: a near-verbatim drift-cohort prompt failing to appear in search results at all was only recognizable as broken because we knew, with certainty, that it *should* have matched.
- **Empirical sigma tuning**: comparing detector output against known `drift` labels directly, rather than guessing whether a threshold "seemed" reasonable.
- **Validating the multi-cluster baseline change**: a before/after recall comparison is only meaningful against a fixed, known-correct test set.

Related Pages

- Sigma Threshold Tuning Methodology
- Multi-Cluster Baseline (41%100%)
- Automated Validation / Acceptance Testing